

SECTION 1: Identification of the substance/mixture and of the company/undertaking**1.1 Product identifier**

Product name	SAF-PD
UFI:	2D73-U0N3-A00M-3E7P
Product code	454252-DE01
SDS #	454252
Product type	Liquid.

1.2 Relevant identified uses of the substance or mixture and uses advised against

Identified uses
General use of lubricants and greases in vehicles or machinery-Industrial General use of lubricants and greases in vehicles or machinery-Professional

Use of the substance/ mixture	Gear lubricant For specific application advice see appropriate Technical Data Sheet or consult our company representative.
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1.3 Details of the supplier of the safety data sheet

Supplier	Castrol Holdings Europe B.V., d'Arcyweg 76, 3198NA Europoort Rotterdam Castrol Germany GmbH, Überseeallee 1, 20457 Hamburg +49 (0) 800 863 73 70 E-mail address MSDSadvice@bp.com
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1.4 Emergency telephone number

EMERGENCY TELEPHONE NUMBER	Carechem: +44 (0) 1235 239 670 (24/7)
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SECTION 2: Hazards identification**2.1 Classification of the substance or mixture**

Product definition	Mixture
Classification according to Regulation (EC) No. 1272/2008 [CLP/GHS]	

Skin Sens. 1, H317

Aquatic Chronic 2, H411

See Section 16 for the full text of the H statements declared above.

See sections 11 and 12 for more detailed information on health effects and symptoms and environmental hazards.

2.2 Label elements

UFI:	2D73-U0N3-A00M-3E7P
Hazard pictograms	



Signal word	Warning
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Hazard statements	H317 - May cause an allergic skin reaction. H411 - Toxic to aquatic life with long lasting effects.
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Precautionary statements

General	P102 - Keep out of reach of children. P101 - If medical advice is needed, have product container or label at hand.
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Product name SAF-PD	Product code 454252-DE01	Page: 1/28
Version 19	Date of issue 27 November 2025	Format Germany
Date of previous issue 20 February 2024.		Language ENGLISH (Germany)

SECTION 2: Hazards identification

Prevention	P280 - Wear protective gloves. P273 - Avoid release to the environment. P261 - Avoid breathing vapour.
Response	P391 - Collect spillage. P302 + P352 - IF ON SKIN: Wash with plenty of soap and water. P333 + P313 - If skin irritation or rash occurs: Get medical attention. P362 + P364 - Take off contaminated clothing and wash it before reuse.
Storage	Not applicable.
Disposal	P501 - Dispose of contents and container in accordance with all local, regional, national and international regulations.
Hazardous ingredients	2,6-di-tert-butyl-4-nonylphenol Fatty acids, C16-18 (even numbered, C18 unsaturated), 2-ethylhexyl esters, epoxidized Molybdenum trioxide, reaction products with bis[O,O-bis(2-ethylhexyl)] hydrogen dithiophosphate Reaction products of triphenyl phosphite and isodecanol (1:1)
Supplemental label elements	Not applicable.

EU Regulation (EC) No. 1907/2006 (REACH)

Annex XVII - Restrictions on the manufacture, placing on the market and use of certain dangerous substances, mixtures and articles	Not applicable.
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Special packaging requirements

Containers to be fitted with child-resistant fastenings	Not applicable.
Tactile warning of danger	Not applicable.

2.3 Other hazards

Product meets the criteria for PBT or vPvB according to Regulation (EC) No. 1907/2006, Annex XIII	This mixture does not contain any substances that are assessed to be a PBT or a vPvB.
Product meets the criteria for endocrine disrupting properties according to Regulation (EC) No. 1907/2006.	This substance/mixture does not contain any components that are considered to have endocrine disrupting properties.
Other hazards which do not result in classification	Defatting to the skin.

SECTION 3: Composition/information on ingredients

3.2 Mixtures

Product definition	Mixture
Synthetic base stock. Proprietary performance additives.	

Product/ingredient name	Identifiers	%	Classification	Specific Conc. Limits, M-factors and ATEs	Type
2,6-di-tert-butyl-4-nonylphenol	REACH #: 01-2120759723-46 EC: 224-320-7 CAS: 4306-88-1	≤3	Skin Sens. 1B, H317 Aquatic Acute 1, H400 Aquatic Chronic 1, H410	M [Acute] = 1 M [Chronic] = 1	[1]
Polysulfides, di-tert-Bu	REACH #: 01-2119540515-43 EC: 273-103-3 CAS: 68937-96-2	≤3	Skin Sens. 1B, H317 Aquatic Chronic 3, H412	Skin Sens. 1, H317: C ≥ 46%	[1]
Distillates (petroleum), hydrotreated heavy paraffinic	REACH #: 01-2119484627-25 EC: 265-157-1 CAS: 64742-54-7 Index: 649-467-00-8	≤3	Asp. Tox. 1, H304	-	[1]

Product name	SAF-PD	Product code	454252-DE01	Page:	2/28
Version	19	Date of issue	27 November 2025	Format	Germany (Germany)
Date of previous issue	20 February 2024.			Language	ENGLISH

SECTION 3: Composition/information on ingredients

Distillates (petroleum), solvent-dewaxed heavy paraffinic	REACH #: 01-2119471299-27 EC: 265-169-7 CAS: 64742-65-0 Index: 649-474-00-6	≤3	Asp. Tox. 1, H304	-	[1]
Fatty acids, C16-18 (even numbered, C18 unsaturated), 2-ethylhexyl esters, epoxidized	REACH #: 01-2119977115-34 EC: 701-432-9 CAS: -	≤3	Skin Sens. 1B, H317	-	[1]
Zinc bis[O,O-bis(2-ethylhexyl)] bis(dithiophosphate)	REACH #: 01-2119493635-27 EC: 224-235-5 CAS: 4259-15-8	≤3	Eye Dam. 1, H318 Aquatic Chronic 2, H411	Eye Dam. 1, H318: C ≥ 50%	[1]
Molybdenum trioxide, reaction products with bis[O,O-bis(2-ethylhexyl)] hydrogen dithiophosphate	REACH #: 01-2120772600-59 EC: 947-946-9 CAS: -	<1	Skin Irrit. 2, H315 Skin Sens. 1B, H317 Aquatic Chronic 4, H413	-	[1]
Phenol, isopropylated, phosphate (3:1) [Triphenyl phosphate >5%]	REACH #: 01-2119535109-41 EC: 273-066-3 CAS: 68937-41-7	≤1	Repr. 2, H361fd STOT RE 2, H373 Aquatic Chronic 1, H410	M [Chronic] = 10	[1] [2]
Benzenamine, N-phenyl-, reaction products with 2,4,4-trimethylpentene	REACH #: 01-2119491299-23 EC: 270-128-1 CAS: 68411-46-1	≤1	Repr. 2, H361f	-	[1]
Reaction products of triphenyl phosphite and isodecanol (1:1)	REACH #: 01-2119968254-31 EC: 701-341-4 CAS: -	<1	Skin Sens. 1, H317 STOT RE 2, H373 Aquatic Chronic 2, H411	-	[1]

See Section 16 for the full text of the H statements declared above.

Type

- [1] Substance classified with a health or environmental hazard
[2] Substance with a workplace exposure limit
Occupational exposure limits, if available, are listed in Section 8.

SECTION 4: First aid measures

4.1 Description of first aid measures

Eye contact	In case of contact, immediately flush eyes with plenty of water for at least 15 minutes. Eyelids should be held away from the eyeball to ensure thorough rinsing. Check for and remove any contact lenses. Get medical attention.
Skin contact	Wash skin thoroughly with soap and water or use recognised skin cleanser. Remove contaminated clothing and shoes. Wash clothing before reuse. Clean shoes thoroughly before reuse. In the event of any complaints or symptoms, avoid further exposure. Get medical attention.
Inhalation	If inhaled, remove to fresh air. Get medical attention if symptoms occur.
Ingestion	Do not induce vomiting unless directed to do so by medical personnel. Never give anything by mouth to an unconscious person. If unconscious, place in recovery position and get medical attention immediately. Get medical attention if symptoms occur.
Protection of first-aiders	No action shall be taken involving any personal risk or without suitable training. It may be dangerous to the person providing aid to give mouth-to-mouth resuscitation. Wash contaminated clothing thoroughly with water before removing it, or wear gloves.

4.2 Most important symptoms and effects, both acute and delayed

See Section 11 for more detailed information on health effects and symptoms.

Potential acute health effects

Inhalation	Vapour inhalation under ambient conditions is not normally a problem due to low vapour pressure.
Ingestion	No known significant effects or critical hazards.
Skin contact	Defatting to the skin. May cause skin dryness and irritation. May cause an allergic skin reaction.
Eye contact	See: Section 11. Toxicological Information - Potential acute health effects: Eye contact

Delayed and immediate effects as well as chronic effects from short and long-term exposure

Inhalation	Overexposure to the inhalation of airborne droplets or aerosols may cause irritation of the respiratory tract.
Ingestion	Ingestion of large quantities may cause nausea and diarrhoea.

Product name	SAF-PD	Product code	454252-DE01	Page:	3/28
Version	19	Date of issue	27 November 2025	Format	Germany
Date of previous issue	20 February 2024.			Language	ENGLISH
			(Germany)		

SECTION 4: First aid measures

Skin contact	Prolonged or repeated contact can defat the skin and lead to irritation and/or dermatitis.
Eye contact	Potential risk of transient stinging or redness if accidental eye contact occurs.

4.3 Indication of any immediate medical attention and special treatment needed

Notes to physician	Treatment should in general be symptomatic and directed to relieving any effects.
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SECTION 5: Firefighting measures

5.1 Extinguishing media

Suitable extinguishing media	Use foam or all-purpose dry chemical to extinguish.
Unsuitable extinguishing media	Do not use water jet. The use of a water jet may cause the fire to spread by splashing the burning product.

5.2 Special hazards arising from the substance or mixture

Hazards from the substance or mixture	In a fire or if heated, a pressure increase will occur and the container may burst.
Hazardous combustion products	 Combustion products may include the following: carbon oxides (CO, CO ₂) (carbon monoxide, carbon dioxide) metal oxide/oxides phosphorus oxides sulphur oxides (SO, SO ₂ , etc.)

5.3 Advice for firefighters

Special precautions for fire-fighters	No action shall be taken involving any personal risk or without suitable training. Promptly isolate the scene by removing all persons from the vicinity of the incident if there is a fire. This material is toxic to aquatic organisms. Fire water contaminated with this material must be contained and prevented from being discharged to any waterway, sewer or drain.
Special protective equipment for fire-fighters	Fire-fighters should wear appropriate protective equipment and self-contained breathing apparatus (SCBA) with a full face-piece operated in positive pressure mode. Clothing for fire-fighters (including helmets, protective boots and gloves) conforming to European standard EN 469 will provide a basic level of protection for chemical incidents.

SECTION 6: Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures

For non-emergency personnel	Contact emergency personnel. No action shall be taken involving any personal risk or without suitable training. Evacuate surrounding areas. Keep unnecessary and unprotected personnel from entering. Do not touch or walk through spilt material. Floors may be slippery; use care to avoid falling. Avoid breathing vapour or mist. Provide adequate ventilation. Put on appropriate personal protective equipment.
For emergency responders	Entry into a confined space or poorly ventilated area contaminated with vapour, mist or fume is extremely hazardous without the correct respiratory protective equipment and a safe system of work. Wear self-contained breathing apparatus. Wear a suitable chemical protective suit. Chemical resistant boots. See also the information in "For non-emergency personnel".

6.2 Environmental precautions

Avoid dispersal of spilt material and runoff and contact with soil, waterways, drains and sewers. Inform the relevant authorities if the product has caused environmental pollution (sewers, waterways, soil or air). Water polluting material. May be harmful to the environment if released in large quantities. Collect spillage.

6.3 Methods and material for containment and cleaning up

Small spill	Stop leak if without risk. Move containers from spill area. Absorb with an inert material and place in an appropriate waste disposal container. Dispose of via a licensed waste disposal contractor.
Large spill	Stop leak if without risk. Move containers from spill area. Approach the release from upwind. Prevent entry into sewers, water courses, basements or confined areas. Contain and collect spillage with non-combustible, absorbent material e.g. sand, earth, vermiculite or diatomaceous earth and place in container for disposal according to local regulations. Contaminated absorbent material may pose the same hazard as the spilt product. Dispose of via a licensed waste disposal contractor.

Product name		SAF-PD		Product code		454252-DE01		Page: 4/28	
Version		19		Date of issue		27 November 2025		Format	
								Germany	
								Language	
								ENGLISH	
Date of previous issue									

SECTION 6: Accidental release measures

6.4 Reference to other sections	See Section 1 for emergency contact information. See Section 5 for firefighting measures. See Section 8 for information on appropriate personal protective equipment. See Section 12 for environmental precautions. See Section 13 for additional waste treatment information.
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SECTION 7: Handling and storage

The information in this section contains generic advice and guidance. The list of Identified Uses in Section 1 should be consulted for any available use-specific information provided in the Exposure Scenario(s).

7.1 Precautions for safe handling

Protective measures	Put on appropriate personal protective equipment. Persons with a history of skin sensitization problems should not be employed in any process in which this product is used. Do not get in eyes or on skin or clothing. Do not ingest. Avoid breathing vapour or mist. Avoid contact of spilt material and runoff with soil and surface waterways. Keep in the original container or an approved alternative made from a compatible material, kept tightly closed when not in use. Do not reuse container. Empty containers retain product residue and can be hazardous.
Advice on general occupational hygiene	Eating, drinking and smoking should be prohibited in areas where this material is handled, stored and processed. Wash thoroughly after handling. Remove contaminated clothing and protective equipment before entering eating areas. See also Section 8 for additional information on hygiene measures.

7.2 Conditions for safe storage, including any incompatibilities

	Store in accordance with local regulations. Store in a dry, cool and well-ventilated area, away from incompatible materials (see Section 10). Keep away from heat and direct sunlight. Keep container tightly closed and sealed until ready for use. Containers that have been opened must be carefully resealed and kept upright to prevent leakage. Store and use only in equipment/containers designed for use with this product. Do not store in unlabelled containers. Use appropriate containment to avoid environmental contamination.
Not suitable	Prolonged exposure to elevated temperature.
Germany - Storage code	10

7.3 Specific end use(s)

Recommendations	See section 1.2 and Exposure scenarios in annex, if applicable.
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SECTION 8: Exposure controls/personal protection

The information in this section contains generic advice and guidance. The list of Identified Uses in Section 1 should be consulted for any available use-specific information provided in the Exposure Scenario(s).

8.1 Control parameters

Occupational exposure limits

Product/ingredient name	Exposure limit values
Phenol, isopropylated, phosphate (3:1) [Triphenyl phosphate >5%]	TRGS 900 OEL (Germany) PEAK 15 minutes: 2 mg/m³. Form: Inhalable fraction. Issued/Revised: 5/2016. TWA 8 hours: 1 mg/m³. Form: Inhalable fraction. Issued/Revised: 5/2016.

Whilst specific OELs for certain components may be shown in this section, other components may be present in any mist, vapour or dust produced. Therefore, the specific OELs may not be applicable to the product as a whole and are provided for guidance only.

Recommended monitoring procedures	Reference should be made to monitoring standards, such as the following: European Standard EN 689 (Workplace atmospheres - Guidance for the assessment of exposure by inhalation to chemical agents for comparison with limit values and measurement strategy) European Standard EN 14042 (Workplace atmospheres - Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents) European Standard EN 482 (Workplace atmospheres - General requirements for the performance of procedures for the measurement of chemical agents) Reference to national guidance documents for methods for the determination of hazardous substances will also be required.
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Biological exposure indices

Product/ingredient name	Exposure indices
Molybdenum trioxide, reaction products with bis[O, O-bis(2-ethylhexyl)] hydrogen dithiophosphate	DFG BEI-values list (Germany, 7/2024) [Molybdenum and its compounds] BEI: See Section XII.2: Substances for which no BAT values are currently be derived, but documentaries in the "work Medico-toxicological justifications for BAT values, EKA and BLW", molybdenum [in urine, in plasma]. BEI: 150 µg/l, molybdenum [in urine].

Product name	SAF-PD	Product code	454252-DE01	Page:	5/28
Version	19	Date of issue	27 November 2025	Format	Germany
Date of previous issue	20 February 2024.			Language	ENGLISH
					(Germany)

SECTION 8: Exposure controls/personal protection

DNELs/DMELs

Not available.

PNECs

Not available.

8.2 Exposure controls

Appropriate engineering controls

Provide exhaust ventilation or other engineering controls to keep the relevant airborne concentrations below their respective occupational exposure limits.
All activities involving chemicals should be assessed for their risks to health, to ensure exposures are adequately controlled. Personal protective equipment should only be considered after other forms of control measures (e.g. engineering controls) have been suitably evaluated. Personal protective equipment should conform to appropriate standards, be suitable for use, be kept in good condition and properly maintained.
Your supplier of personal protective equipment should be consulted for advice on selection and appropriate standards. For further information contact your national organisation for standards. The final choice of protective equipment will depend upon a risk assessment. It is important to ensure that all items of personal protective equipment are compatible.

Individual protection measures

Hygiene measures

Wash hands, forearms and face thoroughly after handling chemical products, before eating, smoking and using the lavatory and at the end of the working period. Appropriate techniques should be used to remove potentially contaminated clothing. Wash contaminated clothing before reusing. Ensure that eyewash stations and safety showers are close to the workstation location.

Respiratory protection

In case of insufficient ventilation, wear suitable respiratory equipment.
The correct choice of respiratory protection depends upon the chemicals being handled, the conditions of work and use, and the condition of the respiratory equipment. Safety procedures should be developed for each intended application. Respiratory protection equipment should therefore be chosen in consultation with the supplier/manufacturer and with a full assessment of the working conditions.

Eye/face protection

Safety glasses with side shields.

Skin protection

Hand protection

General Information:

Because specific work environments and material handling practices vary, safety procedures should be developed for each intended application. The correct choice of protective gloves depends upon the chemicals being handled, and the conditions of work and use. Most gloves provide protection for only a limited time before they must be discarded and replaced (even the best chemically resistant gloves will break down after repeated chemical exposures).

Gloves should be chosen in consultation with the supplier / manufacturer and taking account of a full assessment of the working conditions.

Recommended: Nitrile gloves.

Breakthrough time:

Breakthrough time data are generated by glove manufacturers under laboratory test conditions and represent how long a glove can be expected to provide effective permeation resistance. It is important when following breakthrough time recommendations that actual workplace conditions are taken into account. Always consult with your glove supplier for up-to-date technical information on breakthrough times for the recommended glove type.
Our recommendations on the selection of gloves are as follows:

Continuous contact:

Gloves with a minimum breakthrough time of 240 minutes, or >480 minutes if suitable gloves can be obtained.
If suitable gloves are not available to offer that level of protection, gloves with shorter breakthrough times may be acceptable as long as appropriate glove maintenance and replacement regimes are determined and adhered to.

Short-term / splash protection:

Recommended breakthrough times as above.
It is recognised that for short-term, transient exposures, gloves with shorter breakthrough times may commonly be used. Therefore, appropriate maintenance and replacement regimes must be determined and rigorously followed.

SECTION 8: Exposure controls/personal protection

Glove Thickness:

For general applications, we recommend gloves with a thickness typically greater than 0.35 mm.

It should be emphasised that glove thickness is not necessarily a good predictor of glove resistance to a specific chemical, as the permeation efficiency of the glove will be dependent on the exact composition of the glove material. Therefore, glove selection should also be based on consideration of the task requirements and knowledge of breakthrough times. Glove thickness may also vary depending on the glove manufacturer, the glove type and the glove model. Therefore, the manufacturers' technical data should always be taken into account to ensure selection of the most appropriate glove for the task.

Note: Depending on the activity being conducted, gloves of varying thickness may be required for specific tasks. For example:

- Thinner gloves (down to 0.1 mm or less) may be required where a high degree of manual dexterity is needed. However, these gloves are only likely to give short duration protection and would normally be just for single use applications, then disposed of.
- Thicker gloves (up to 3 mm or more) may be required where there is a mechanical (as well as a chemical) risk i.e. where there is abrasion or puncture potential.

Skin and body

Use of protective clothing is good industrial practice. Personal protective equipment for the body should be selected based on the task being performed and the risks involved and should be approved by a specialist before handling this product. Cotton or polyester/cotton overalls will only provide protection against light superficial contamination that will not soak through to the skin. Overalls should be laundered on a regular basis. When the risk of skin exposure is high (e.g. when cleaning up spillages or if there is a risk of splashing) then chemical resistant aprons and/or impervious chemical suits and boots will be required.

Refer to standards:

- Respiratory protection: EN 529
- Gloves: EN 420, EN 374
- Eye protection: EN 166
- Filtering half-mask: EN 149
- Filtering half-mask with valve: EN 405
- Half-mask: EN 140 plus filter
- Full-face mask: EN 136 plus filter
- Particulate filters: EN 143
- Gas/combined filters: EN 14387

Environmental exposure controls

Emissions from ventilation or work process equipment should be checked to ensure they comply with the requirements of environmental protection legislation. In some cases, fume scrubbers, filters or engineering modifications to the process equipment will be necessary to reduce emissions to acceptable levels.

SECTION 9: Physical and chemical properties

The conditions of measurement of all properties are at standard temperature and pressure unless otherwise indicated.

9.1 Information on basic physical and chemical properties

Physical state	Liquid.
Colour	Green.
Odour	Not available.
Odour threshold	Not available.
Melting point/freezing point	Not available.
Initial boiling point and boiling range	Not available.
Flammability	Not available.
Lower and upper explosion limit	Not available.
Flash point	Open cup: >200°C (>392°F) [Cleveland ASTM D 92]
Auto-ignition temperature	

SECTION 9: Physical and chemical properties

	Ingredient name	°C	°F	Method
Decomposition temperature	Dec-1-ene, homopolymer, hydrogenated Dec-1-ene, oligomers, hydrogenated	343 to 369	649.4 to 696.2	ASTM D 2159
	Dec-1-ene, homopolymer, hydrogenated Dec-1-ene, oligomers, hydrogenated	343 to 369	649.4 to 696.2	ASTM D 2159
pH	Not available.			
Kinematic viscosity	Not applicable.			
Solubility	Kinematic: 71.2 mm²/s (71.2 cSt) at 40°C			
	Kinematic: 11 to 12 mm²/s (11 to 12 cSt) at 100°C			
Media		Result		
water		Not soluble		

Partition coefficient n-octanol/ water (log value)

Not applicable.

Vapour pressure	Ingredient name	Vapour Pressure at 20°C		Vapour pressure at 50°C		
		mm Hg	kPa	Method	mm Hg	kPa
Density and/or Relative density	Dec-1-ene, homopolymer, hydrogenated Dec-1-ene, oligomers, hydrogenated	<0.0041	<0.00055	ASTM E 1194-87		
	Polysulfides, di-tert-Bu	0.12	0.016	OECD 104		
	Distillates (petroleum), hydrotreated heavy naphthenic	<0.07501	<0.01	ASTM D 5191		
	Distillates (petroleum), hydrotreated heavy paraffinic	<0.07501	<0.01	ASTM D 5191		
	Distillates (petroleum), solvent-dewaxed heavy paraffinic	<0.07501	<0.01	ASTM D 5191		

Density and/or Relative density

<1000 kg/m³ (<1 g/cm³) at 15°C

Relative vapour density

Not available.

Particle characteristics

Median particle size

Not applicable.

9.2 Other information

Evaporation rate

Not available.

Explosive properties

Not available.

Oxidising properties

Not available.

SECTION 10: Stability and reactivity

10.1 Reactivity

No specific test data available for this product. Refer to Conditions to avoid and Incompatible materials for additional information.

10.2 Chemical stability

The product is stable.

10.3 Possibility of hazardous reactions

Under normal conditions of storage and use, hazardous reactions will not occur.

Under normal conditions of storage and use, hazardous polymerisation will not occur.

10.4 Conditions to avoid

Avoid all possible sources of ignition (spark or flame).

10.5 Incompatible materials

Reactive or incompatible with the following materials: oxidising materials.

10.6 Hazardous decomposition products

Under normal conditions of storage and use, hazardous decomposition products should not be produced.

SECTION 11: Toxicological information

11.1 Information on hazard classes as defined in Regulation (EC) No 1272/2008

Acute toxicity

Product/ingredient name	Result
2,6-di-tert-butyl-4-nonylphenol	Rat - Oral - LD50 >2000 mg/kg OECD 401
	Rat - Dermal - LD50 >2000 mg/kg OECD 402
Distillates (petroleum), hydrotreated heavy paraffinic	Rat - Oral - LD50 >5000 mg/kg OECD 401
	Rabbit - Dermal - LD50 >5000 mg/kg OECD 402
	Rat - Inhalation - LC50 Dusts and mists >5 mg/l [4 hours] OECD 403
Distillates (petroleum), solvent-dewaxed heavy paraffinic	Rat - Oral - LD50 >5000 mg/kg OECD 401
	Rat - Dermal - LD50 >2000 mg/kg OECD 402
	Rat - Inhalation - LC50 Dusts and mists >5.53 mg/l [4 hours] OECD 403
zinc bis[O,O-bis(2-ethylhexyl)] bis (dithiophosphate)	Rat - Oral - LD50 3100 mg/kg OECD 401
	Rat - Dermal - LD50 >5000 mg/kg OECD 402
Molybdenum trioxide, reaction products with bis [O,O-bis(2-ethylhexyl)] hydrogen dithiophosphate	Rat - Oral - LD50 6015 mg/kg Equivalent to OECD 401
	Rabbit - Dermal - LD50 8833 mg/kg Equivalent to OECD 402
Benzenamine, N-phenyl-, reaction products with 2,4,4-trimethylpentene	Rat - Oral - LC50 >5000 mg/kg OECD 401
	Rat - Dermal - LC50 >2000 mg/kg OECD 402
Reaction products of triphenyl phosphite and isodecanol (1:1)	Rat - Oral - LC50 3840 mg/kg OECD 401
	Rabbit - Dermal - LC50 >5000 mg/kg OECD 402
	Rat - Inhalation - LC50 Vapour >8.4 mg/l [4 hours] OECD 403

SECTION 11: Toxicological information

Acute toxicity estimates

Product/ingredient name	Oral (mg/kg)	Dermal (mg/kg)	Inhalation (gases) (ppm)	Inhalation (vapours) (mg/l)	Inhalation (dusts and mists) (mg/l)
zinc bis[O,O-bis(2-ethylhexyl)] bis (dithiophosphate)	2500	N/A	N/A	N/A	N/A
Reaction products of triphenyl phosphite and isodecanol (1:1)	2500	N/A	N/A	N/A	N/A

Skin corrosion/irritation

Product/ingredient name	Result
2,6-di-tert-butyl-4-nonylphenol	Rabbit - Skin - Mild irritant OECD 404
Distillates (petroleum), hydrotreated heavy paraffinic	Rabbit - Skin - Mild irritant OECD 404
Distillates (petroleum), solvent-dewaxed heavy paraffinic	Rabbit - Skin - Non-irritant to skin.
zinc bis[O,O-bis(2-ethylhexyl)] bis (dithiophosphate)	Rabbit - Skin - Non-irritant to skin. Duration of treatment/exposure: 4 hours
Molybdenum trioxide, reaction products with bis [O,O-bis(2-ethylhexyl)] hydrogen dithiophosphate	Unspecified - Skin - Irritant OECD 439
Benzenamine, N-phenyl-, reaction products with 2,4,4-trimethylpentene	Rabbit - Skin - Slightly irritating to the skin. OECD 404
Reaction products of triphenyl phosphite and isodecanol (1:1)	Rabbit - Skin - Slightly irritating to the skin. OECD 404

Serious eye damage/eye irritation

Product/ingredient name	Result
2,6-di-tert-butyl-4-nonylphenol	Rabbit - Eyes - Slightly irritating to the eyes. OECD 405
Distillates (petroleum), hydrotreated heavy paraffinic	Rabbit - Eyes - Non-irritating to the eyes. OECD 405
Distillates (petroleum), solvent-dewaxed heavy paraffinic	Rabbit - Eyes - Non-irritating to the eyes. OECD 405
Molybdenum trioxide, reaction products with bis [O,O-bis(2-ethylhexyl)] hydrogen dithiophosphate	Unspecified - Eyes - Non-irritating to the eyes. OECD 437
Benzenamine, N-phenyl-, reaction products with 2,4,4-trimethylpentene	Rabbit - Eyes - Not irritant OECD 405
Reaction products of triphenyl phosphite and isodecanol (1:1)	Rabbit - Eyes - Slightly irritating to the eyes. ASTM

Respiratory corrosion/irritation

Not available.

Respiratory or skin sensitization

Product/ingredient name	Result
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SECTION 11: Toxicological information

 2,6-di-tert-butyl-4-nonylphenol	Mouse - skin OECD 429 <u>Result:</u> Sensitising
Distillates (petroleum), hydrotreated heavy paraffinic	Guinea pig - skin OECD 406 <u>Result:</u> Not sensitising
Distillates (petroleum), solvent-dewaxed heavy paraffinic	Guinea pig - skin OECD 406 <u>Result:</u> Not sensitising
zinc bis[O,O-bis(2-ethylhexyl)] bis (dithiophosphate)	Guinea pig - skin OECD 406 <u>Result:</u> Not sensitising
Molybdenum trioxide, reaction products with bis [O,O-bis(2-ethylhexyl)] hydrogen dithiophosphate	Mouse - skin OECD 429 <u>Result:</u> Sensitising
Benzenamine, N-phenyl-, reaction products with 2,4,4-trimethylpentene	Guinea pig - skin OECD 406 <u>Result:</u> Not sensitising
Reaction products of triphenyl phosphite and isodecanol (1:1)	Guinea pig - skin OECD 406 <u>Result:</u> Sensitising

Germ cell mutagenicity

Product/ingredient name	Result
 2,6-di-tert-butyl-4-nonylphenol	In vitro - Bacteria OECD 471 <u>Result:</u> Negative In vitro - Mammalian-Animal OECD 487 <u>Result:</u> Negative In vitro - Mammalian-Animal OECD 4776 <u>Result:</u> Negative
Distillates (petroleum), hydrotreated heavy paraffinic	In vitro - Bacteria Bacterial Reverse Mutation Test <u>Result:</u> Negative In vitro - Mammal - species unspecified In vitro Mammalian Chromosomal Aberration Test <u>Result:</u> Negative In vivo - Mammal - species unspecified Mammalian Erythrocyte Micronucleus Test <u>Result:</u> Negative
Distillates (petroleum), solvent-dewaxed heavy paraffinic	In vitro - Mammal - species unspecified In vitro Mammalian Cell Gene Mutation Test <u>Result:</u> Negative In vitro - Bacteria OECD [Bacterial Reverse Mutation Test] <u>Result:</u> Negative In vitro - Mammal - species unspecified OECD [In vitro Mammalian Chromosomal Aberration Test] <u>Result:</u> Negative
Molybdenum trioxide, reaction products with bis [O,O-bis(2-ethylhexyl)] hydrogen dithiophosphate	In vitro - Bacteria OECD [Bacterial Reverse Mutation Test] <u>Result:</u> Negative

SECTION 11: Toxicological information

Benzenamine, N-phenyl-, reaction products with 2,4,4-trimethylpentene	In vitro - Mammal - species unspecified OECD [In vitro Micronucleus Test] <u>Result:</u> Negative
	In vitro - Mammal - species unspecified OECD 490 <u>Result:</u> Negative
	In vitro - Bacteria OECD 471 <u>Result:</u> Negative
	In vitro - Mammalian-Animal OECD 487 <u>Result:</u> Negative
Reaction products of triphenyl phosphite and isodecanol (1:1)	In vitro - Mammalian-Animal OECD 476 <u>Result:</u> Negative
	In vitro - Bacteria OECD 471 <u>Result:</u> Negative
	In vitro - Mammalian-Animal OECD 487 <u>Result:</u> Negative
	In vivo - Mammalian-Animal OECD 474 <u>Result:</u> Negative

Carcinogenicity

Product/ingredient name	Result
Distillates (petroleum), hydrotreated heavy paraffinic	Mouse - Dermal - Unspecified OECD 451 <u>Result:</u> Negative
Distillates (petroleum), solvent-dewaxed heavy paraffinic	Mouse - Dermal - Unspecified OECD 451 <u>Result:</u> Negative

Reproductive toxicity

Product/ingredient name	Result
2,6-di-tert-butyl-4-nonylphenol	Rat - Oral OECD 422 <u>Maternal toxicity:</u> Positive <u>Fertility effects:</u> Negative
Distillates (petroleum), hydrotreated heavy paraffinic	Rat - Oral OECD 421 <u>Maternal toxicity:</u> Negative <u>Fertility effects:</u> Negative <u>Developmental:</u> Negative
Distillates (petroleum), solvent-dewaxed heavy paraffinic	Rat - Oral OECD 421 <u>Maternal toxicity:</u> Negative <u>Fertility effects:</u> Negative <u>Developmental:</u> Negative
Molybdenum trioxide, reaction products with bis [O,O-bis(2-ethylhexyl)] hydrogen dithiophosphate	Rat - Oral OECD 422 <u>Maternal toxicity:</u> Negative <u>Fertility effects:</u> Negative <u>Developmental:</u> Negative

SECTION 11: Toxicological information

Benzenamine, N-phenyl-, reaction products with 2,4,4-trimethylpentene	Rat - Oral OECD 443 <u>Maternal toxicity</u> : Negative <u>Fertility effects</u> : Positive <u>Developmental</u> : Negative
Reaction products of triphenyl phosphite and isodecanol (1:1)	Rat - Oral OECD 422 <u>Maternal toxicity</u> : Negative <u>Fertility effects</u> : Negative <u>Developmental</u> : Negative

Specific target organ toxicity (single exposure)

Not available.

Specific target organ toxicity (repeated exposure)

Product/ingredient name	Result
Phenol, isopropylated, phosphate (3:1)	STOT RE 2, H373
Reaction products of triphenyl phosphite and isodecanol (1:1)	STOT RE 2, H373

Aspiration hazard

Product/ingredient name	Result
Distillates (petroleum), hydrotreated heavy paraffinic	ASPIRATION HAZARD - Category 1
Distillates (petroleum), solvent-dewaxed heavy paraffinic	ASPIRATION HAZARD - Category 1

Information on likely routes of exposure

Routes of entry anticipated: Oral, Dermal, Inhalation, Eyes.

Potential acute health effects

Inhalation	Vapour inhalation under ambient conditions is not normally a problem due to low vapour pressure.
Ingestion	No known significant effects or critical hazards.
Skin contact	Defatting to the skin. May cause skin dryness and irritation. May cause an allergic skin reaction.
Eye contact	No known significant effects or critical hazards.

Symptoms related to the physical, chemical and toxicological characteristics

Inhalation	May be harmful by inhalation if exposure to vapour, mists or fumes resulting from thermal decomposition products occurs.
Ingestion	No specific data.
Skin contact	Adverse symptoms may include the following: irritation redness dryness cracking
Eye contact	No specific data.

Delayed and immediate effects as well as chronic effects from short and long-term exposure

Inhalation	Overexposure to the inhalation of airborne droplets or aerosols may cause irritation of the respiratory tract.
Ingestion	Ingestion of large quantities may cause nausea and diarrhoea.
Skin contact	Prolonged or repeated contact can defat the skin and lead to irritation and/or dermatitis.
Eye contact	Potential risk of transient stinging or redness if accidental eye contact occurs.

Potential chronic health effects

Not available.

Conclusion/Summary [Product]	Not available.
General	No known significant effects or critical hazards.
Carcinogenicity	No known significant effects or critical hazards.
Mutagenicity	No known significant effects or critical hazards.
Developmental effects	No known significant effects or critical hazards.
Fertility effects	No known significant effects or critical hazards.

SECTION 11: Toxicological information

11.2 Information on other hazards

11.2.1 Endocrine disrupting properties

Conclusion/Summary [Product]

This substance/mixture does not contain any components that are considered to have endocrine disrupting properties.

11.2.2 Other information

Not available.

SECTION 12: Ecological information

12.1 Toxicity

Product/ingredient name	Result
2,6-di-tert-butyl-4-nonylphenol	Acute - ErC50 OECD 201 Algae >100 mg/l [72 hours]
	Acute - EC50 OECD 202 Daphnia 0.124 mg/l [48 hours]
	Acute - LC50 OECD 203 Fish >10 mg/l [96 hours]
	Acute - EC50 OECD 209 Micro-organism >1000 mg/l [3 hours]
	Chronic - EC10 OECD 201 Algae 100 mg/l [72 hours]
Distillates (petroleum), hydrotreated heavy paraffinic	Acute - EL50 OECD 201 Algae >100 mg/l [72 hours]
	Acute - EL50 OECD 202 Daphnia >10000 mg/l [48 hours]
	Acute - LL50 OECD 203 Fish >100 mg/l [96 hours]
	Chronic - NOEL OECD 201 Algae ≥100 mg/l [72 hours]
	Chronic - NOEL OECD 211 Daphnia 10 mg/l [21 days]
Distillates (petroleum), solvent-dewaxed heavy paraffinic	Acute - EL50 OECD 201 Algae >100 mg/l [72 hours]
	Acute - EL50 OECD 202 Daphnia >10000 mg/l [48 hours]

SECTION 12: Ecological information

zinc bis[O,O-bis(2-ethylhexyl)] bis (dithiophosphate)	Acute - LL50 OECD 203 Fish >100 mg/l [96 hours]
	Chronic - NOEL OECD 201 Algae ≥100 mg/l [72 hours]
	Chronic - NOEL OECD 211 Daphnia 10 mg/l [21 days]
	Acute - ErL50 OECD 201 Algae 410 mg/l [72 hours]
	Acute - ErL50 OECD 202 Daphnia 75 mg/l [48 hours]
Molybdenum trioxide, reaction products with bis[O, O-bis(2-ethylhexyl)] hydrogen dithiophosphate	Chronic - NOEC OECD 211 Daphnia 0.4 mg/l [21 days]
	Acute - LL50 OECD 203 Fish 4.4 mg/l [96 hours]
	Acute - EL50 OECD 201 Algae >100 mg/l [72 hours]
	Acute - EL50 OECD 202 Daphnia >100 mg/l [48 hours]
	Acute - LL50 OECD 203 Fish >100 mg/l [96 hours]
Benzenamine, N-phenyl-, reaction products with 2,4,4-trimethylpentene	Chronic - NOEL OECD 201 Algae 100 mg/l [96 hours]
	Acute - ErC50 OECD 201 Algae >100 mg/l [72 hours]
	Acute - EC50 OECD 202 Daphnia 51 mg/l [48 hours]
	Acute - LC50 OECD 203 Fish >100 mg/l [96 hours]
	Chronic - EC10 OECD 211

SECTION 12: Ecological information

	Daphnia 1.69 mg/l [21 days]
	Chronic - NOEC OECD 201 Algae ≥10 mg/l [72 hours]
Reaction products of triphenyl phosphite and isodecanol (1:1)	Acute - EC50 OECD 201 Algae 1.6 mg/l [72 hours]
	Acute - EC50 OECD 202 Daphnia 1 to 5 mg/l [48 hours]
	Acute - LC50 OECD 203 Fish >16 mg/l [96 hours]
	Acute - EC50 OECD 209 Micro-organism >100 mg/l [3 hours]

Environmental hazards Toxic to aquatic life with long lasting effects.

12.2 Persistence and degradability

Not expected to be rapidly degradable.

Product/ingredient name	Result
2,6-di-tert-butyl-4-nonylphenol	OECD 302C 31% [28 days]
Distillates (petroleum), hydrotreated heavy paraffinic	OECD 301F 31% [28 days] - Not readily
Distillates (petroleum), solvent-dewaxed heavy paraffinic	OECD 301F 31% [28 days] - Not readily
zinc bis[O,O-bis(2-ethylhexyl)] bis (dithiophosphate)	<5% [27 days] - Not readily
Molybdenum trioxide, reaction products with bis[O, O-bis(2-ethylhexyl)] hydrogen dithiophosphate	OECD 301B 11% [28 days] - Not readily
Benzenamine, N-phenyl-, reaction products with 2,4,4-trimethylpentene	OECD 301B 1% [28 days]
Reaction products of triphenyl phosphite and isodecanol (1:1)	17% [28 days]

12.3 Bioaccumulative potential

This product is not expected to bioaccumulate through food chains in the environment.

Product/ingredient name	LogP _{ow}	BCF	Potential
Polysulfides, di-tert-Bu	5.6	-	High
zinc bis[O,O-bis (2-ethylhexyl)] bis (dithiophosphate)	3.59	-	Low
Molybdenum trioxide, reaction products with bis[O, O-bis(2-ethylhexyl)] hydrogen dithiophosphate	18.94	-	High
Phenol, isopropylated, phosphate (3:1)	4.92 to 5.17	-	High
Benzenamine, N-phenyl-, reaction products with 2,4,4-trimethylpentene	5.1	-	High

SECTION 12: Ecological information

12.4 Mobility in soil

Soil/water partition coefficient

Product/ingredient name	logKoc	Koc
2,6-di-tert-butyl-4-nonylphenol	4.79	61002.8
Fatty acids, C16-18 (even numbered, C18 unsaturated), 2-ethylhexyl esters, epoxidized	4.01	10197.9
zinc bis[O,O-bis(2-ethylhexyl)] bis (dithiophosphate)	3.46	2912.11
Molybdenum trioxide, reaction products with bis[O,O-bis(2-ethylhexyl)] hydrogen dithiophosphate	3.46	2912.11

Results of PMT and vPvM assessment

Product/ingredient name	PMT	P	M	T	vPvM	vP	vM
2,6-di-tert-butyl-4-nonylphenol	No	No	No	No	No	No	No
Polysulfides, di-tert-Bu	No	No	No	No	No	No	No
Distillates (petroleum), hydrotreated heavy paraffinic	No	No	No	No	No	No	No
Distillates (petroleum), solvent-dewaxed heavy paraffinic	No	No	No	No	No	No	No
Fatty acids, C16-18 (even numbered, C18 unsaturated), 2-ethylhexyl esters, epoxidized	No	No	No	No	No	No	No
zinc bis[O,O-bis(2-ethylhexyl)] bis (dithiophosphate)	No	No	No	No	No	No	No
Molybdenum trioxide, reaction products with bis[O,O-bis(2-ethylhexyl)] hydrogen dithiophosphate	No	No	No	No	No	No	No
Phenol, isopropylated, phosphate (3:1)	No	No	No	No	No	No	No
Benzenamine, N-phenyl-, reaction products with 2,4,4-trimethylpentene	No	No	No	No	No	No	No
Reaction products of triphenyl phosphite and isodecanol (1:1)	No	No	No	No	No	No	No

Mobility Spillages may penetrate the soil causing ground water contamination.

Conclusion/Summary The product does not meet the criteria to be considered as a PMT or vPvM.

12.5 Results of PBT and vPvB assessment

Regulation (EC) No. 1907/2006 [REACH]

Product/ingredient name	PBT	P	B	T	vPvB	vP	vB
2,6-di-tert-butyl-4-nonylphenol	No	N/A	N/A	No	N/A	N/A	N/A
Polysulfides, di-tert-Bu	No	N/A	N/A	No	N/A	N/A	N/A
Distillates (petroleum), hydrotreated heavy paraffinic	No	N/A	N/A	No	N/A	N/A	N/A
Distillates (petroleum), solvent-dewaxed heavy paraffinic	No	N/A	N/A	No	N/A	N/A	N/A
Fatty acids, C16-18 (even numbered, C18 unsaturated), 2-ethylhexyl esters, epoxidized	No	N/A	N/A	No	N/A	N/A	N/A
zinc bis[O,O-bis(2-ethylhexyl)] bis (dithiophosphate)	No	N/A	N/A	No	N/A	N/A	N/A
Molybdenum trioxide, reaction products with bis[O,O-bis(2-ethylhexyl)] hydrogen dithiophosphate	No	N/A	N/A	No	N/A	N/A	N/A
Phenol, isopropylated,	N/A	N/A	N/A	Yes	N/A	N/A	N/A

SECTION 12: Ecological information

phosphate (3:1) Benzenamine, N-phenyl-, reaction products with 2,4,4-trimethylpentene	N/A	N/A	N/A	Yes	N/A	N/A	N/A
Reaction products of triphenyl phosphite and isodecanol (1:1)	N/A	N/A	N/A	Yes	N/A	N/A	N/A

Regulation (EC) No. 1272/2008 [CLP]

Product/ingredient name	PBT	P	B	T	vPvB	vP	vB
2,6-di-tert-butyl- 4-nonylphenol	No	No	No	No	No	No	No
Polysulfides, di-tert-Bu	No	No	No	No	No	No	No
Distillates (petroleum), hydrotreated heavy paraffinic	No	No	No	No	No	No	No
Distillates (petroleum), solvent-dewaxed heavy paraffinic	No	No	No	No	No	No	No
Fatty acids, C16-18 (even numbered, C18 unsaturated), 2-ethylhexyl esters, epoxidized	No	No	No	No	No	No	No
zinc bis[O,O-bis (2-ethylhexyl)] bis (dithiophosphate)	No	No	No	No	No	No	No
Molybdenum trioxide, reaction products with bis[O, O-bis(2-ethylhexyl)]	No	No	No	No	No	No	No
hydrogen dithiophosphate	No	No	No	No	No	No	No
Phenol, isopropylated, phosphate (3:1)	No	No	No	No	No	No	No
Benzenamine, N-phenyl-, reaction products with 2,4,4-trimethylpentene	No	No	No	No	No	No	No
Reaction products of triphenyl phosphite and isodecanol (1:1)	No	No	No	No	No	No	No

Conclusion/Summary The product does not meet the criteria to be considered as a PBT or vPvB.
Regulation (EC) No. 1272/2008
[CLP]

12.6 Endocrine disrupting
properties

Conclusion/Summary [Product] This substance/mixture does not contain any components that are
considered to have endocrine disrupting properties.

Other ecological information Spills may form a film on water surfaces causing physical damage to organisms. Oxygen
transfer could also be impaired.

12.7 Other adverse effects No known significant effects or critical hazards.

SECTION 13: Disposal considerations

The information in this section contains generic advice and guidance. The list of Identified Uses in Section 1 should be consulted
for any available use-specific information provided in the Exposure Scenario(s).

13.1 Waste treatment methods

Product

Methods of disposal Where possible, arrange for product to be recycled. Dispose of via an authorised person/
licensed waste disposal contractor in accordance with local regulations.

Hazardous waste Yes.

European waste catalogue (EWC)

Waste code	Waste designation
13 02 08*	other engine, gear and lubricating oils

However, deviation from the intended use and/or the presence of any potential contaminants may require an alternative waste
disposal code to be assigned by the end user.

Packaging

Product name SAF-PD	Product code 454252-DE01	Page: 18/28
Version 19	Date of issue 27 November 2025	Format Germany
Date of previous issue 20 February 2024.		Language ENGLISH (Germany)

SECTION 13: Disposal considerations

Methods of disposal	Where possible, arrange for product to be recycled. Dispose of via an authorised person/ licensed waste disposal contractor in accordance with local regulations.
Special precautions	This material and its container must be disposed of in a safe way. Care should be taken when handling emptied containers that have not been cleaned or rinsed out. Empty containers or liners may retain some product residues. Empty containers represent a fire hazard as they may contain flammable product residues and vapour. Never weld, solder or braze empty containers. Avoid dispersal of spill material and runoff and contact with soil, waterways, drains and sewers.
References	Commission 2014/955/EU Directive 2008/98/EC

SECTION 14: Transport information

	ADR/RID	ADN	IMDG	IATA
14.1 UN number or ID number	UN3082	UN3082	UN3082	UN3082
14.2 UN proper shipping name	Environmentally hazardous substance, liquid, n.o.s. (2,6-di-tert-butyl-4-nonylphenol, Zinc bis[O,O-bis(2-ethylhexyl)] bis (dithiophosphate))	Environmentally hazardous substance, liquid, n.o.s. (2,6-di-tert-butyl-4-nonylphenol, Zinc bis[O,O-bis (2-ethylhexyl)] bis (dithiophosphate))	Environmentally hazardous substance, liquid, n.o.s.. Marine pollutant (2,6-di-tert-butyl-4-nonylphenol, Zinc bis[O,O-bis (2-ethylhexyl)] bis (dithiophosphate))	Environmentally hazardous substance, liquid, n.o.s. (2,6-di-tert-butyl-4-nonylphenol, Zinc bis[O,O-bis (2-ethylhexyl)] bis (dithiophosphate))
14.3 Transport hazard class(es)	9  	9  	9  	9  
14.4 Packing group	III	III	III	III
14.5 Environmental hazards	Yes.	Yes.	Yes.	Yes.
Additional information	This product is not regulated as a dangerous good when transported in sizes of ≤5 L or ≤5 kg, provided the packagings meet the general provisions of 4.1.1.1, 4.1.1.2 and 4.1.1.4 to 4.1.1.8. Hazard identification number 90 Tunnel code -	This product is not regulated as a dangerous good when transported in sizes of ≤5 L or ≤5 kg, provided the packagings meet the general provisions of 4.1.1.1, 4.1.1.2 and 4.1.1.4 to 4.1.1.8.	This product is not regulated as a dangerous good when transported in sizes of ≤5 L or ≤5 kg, provided the packagings meet the general provisions of 4.1.1.1, 4.1.1.2 and 4.1.1.4 to 4.1.1.8. Emergency schedules F-A, S-F	This product is not regulated as a dangerous good when transported in sizes of ≤5 L or ≤5 kg, provided the packagings meet the general provisions of 5.0.2.4.1, 5.0.2.6.1.1 and 5.0.2.8.

14.6 Special precautions for user Not available.

ADR/RID Classification code: M6

ADN Classification code: M6

14.7 Maritime transport in bulk according to IMO instruments Not available.

SECTION 15: Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

[EU Regulation \(EC\) No. 1907/2006 \(REACH\)](#)

[Annex XIV - List of substances subject to authorisation](#)

[Annex XIV](#)

None of the components are listed.

[Substances of very high concern](#)

Product name	SAF-PD	Product code	454252-DE01	Page:	19/28
Version	19	Date of issue	27 November 2025	Format	Germany
Date of previous issue	20 February 2024.			Language	ENGLISH
			(Germany)		

SECTION 15: Regulatory information

None of the components are listed.

Annex XVII - Restrictions on the manufacture, placing on the market and use of certain dangerous substances, mixtures and articles

Product/ingredient name	%	Designation [Usage]
SAF-PD	95-100	3
4-nonylphenol, branched	<0.01	46
methanol	<0.001	69

Labelling Not applicable.

Other regulations

REACH Status The company, as identified in Section 1, sells this product in the EU in compliance with the current requirements of REACH.

United States inventory (TSCA 8b) All components are active or exempted.

Australia inventory (AIC) All components are listed or exempted.

Canada inventory At least one component is not listed in DSL but all such components are listed in NDSL.

China inventory (IECSC) All components are listed or exempted.

Japan inventory (CSCL) At least one component is not listed.

Korea inventory (KECI) All components are listed or exempted.

Philippines inventory (PICCS) All components are listed or exempted.

Taiwan Chemical Substances Inventory (TCSI) All components are listed or exempted.

Explosive precursors Not applicable.

Ozone depleting substances (EU 2024/590)

Not listed.

Prior Informed Consent (PIC) (649/2012/EU)

Not listed.

Persistent Organic Pollutants

Not listed.

EU - Water framework directive - Priority substances

None of the components are listed.

Seveso Directive

This product is controlled under the Seveso Directive.

Danger criteria

Category
E2

National regulations

Hazardous incident ordinance

This product is controlled under the Germany Hazardous Incident Ordinance.

Danger criteria

Category	Reference number
E2	1.3.2

Hazard class for water 2 (classified according AwSV)

Prohibited Chemicals Regulation (ChemVerbotsV) When placed on the market in Germany, this product is not subject to the Prohibited Chemicals Regulation (ChemVerbotsV).

Occupational restrictions Observe employment restrictions in the following:
Gesetz zum Schutz der arbeitenden Jugend (Jugendarbeitsschutzgesetz – JArbSchG)
Gesetz zum Schutz von Müttern bei der Arbeit, in der Ausbildung und im Studium (Mutterschutzgesetz – MuSchG)

15.2 Chemical safety assessment A Chemical Safety Assessment has been carried out for one or more of the substances within this mixture. A Chemical Safety Assessment has not been carried out for the mixture itself.

SECTION 16: Other information

Abbreviations and acronyms	ADN = European Provisions concerning the International Carriage of Dangerous Goods by Inland Waterway
	ADR = The European Agreement concerning the International Carriage of Dangerous Goods by Road
	ATE = Acute Toxicity Estimate
	BCF = Bioconcentration Factor
	CAS = Chemical Abstracts Service
	CLP = Classification, Labelling and Packaging Regulation [Regulation (EC) No. 1272/2008]
	CSA = Chemical Safety Assessment
	CSR = Chemical Safety Report
	DMEL = Derived Minimal Effect Level
	DNEL = Derived No Effect Level
	EINECS = European Inventory of Existing Commercial chemical Substances
	ES = Exposure Scenario
	EUH statement = CLP-specific Hazard statement
	EWC = European Waste Catalogue
	GHS = Globally Harmonized System of Classification and Labelling of Chemicals
	IATA = International Air Transport Association
	IBC = Intermediate Bulk Container
	IMDG = International Maritime Dangerous Goods
	LogPow = logarithm of the octanol/water partition coefficient
	MARPOL = International Convention for the Prevention of Pollution From Ships, 1973 as modified by the Protocol of 1978. ("Marpol" = marine pollution)
	OECD = Organisation for Economic Co-operation and Development
	PBT = Persistent, Bioaccumulative and Toxic
	PNEC = Predicted No Effect Concentration
	REACH = Registration, Evaluation, Authorisation and Restriction of Chemicals Regulation [Regulation (EC) No. 1907/2006]
	RID = The Regulations concerning the International Carriage of Dangerous Goods by Rail
	RRN = REACH Registration Number
	SADT = Self-Accelerating Decomposition Temperature
	SVHC = Substances of Very High Concern
	STOT-RE = Specific Target Organ Toxicity - Repeated Exposure
	STOT-SE = Specific Target Organ Toxicity - Single Exposure
	TWA = Time weighted average
	UN = United Nations
	UVCB = Complex hydrocarbon substance
	VOC = Volatile Organic Compound
	vPvB = Very Persistent and Very Bioaccumulative
	Varies = may contain one or more of the following 64741-88-4 / RRN 01-2119488706-23, 64741-89-5 / RRN 01-2119487067-30, 64741-95-3 / RRN 01-2119487081-40, 64741-96-4/ RRN 01-2119483621-38, 64742-01-4 / RRN 01-2119488707-21, 64742-44-5 / RRN 01-2119985177-24, 64742-45-6, 64742-52-5 / RRN 01-2119467170-45, 64742-53-6 / RRN 01-2119480375-34, 64742-54-7 / RRN 01-2119484627-25, 64742-55-8 / RRN 01-2119487077-29, 64742-56-9 / RRN 01-2119480132-48, 64742-57-0 / RRN 01-2119489287-22, 64742-58-1, 64742-62-7 / RRN 01-2119480472-38, 64742-63-8, 64742-65-0 / RRN 01-2119471299-27, 64742-70-7 / RRN 01-2119487080-42, 72623-85-9 / RRN 01-2119555262-43, 72623-86-0 / RRN 01-2119474878-16, 72623-87-1 / RRN 01-2119474889-13
Key literature references and sources for data	per reg tab of fusion per default from sub-type - B:

Procedure used to derive the classification according to Regulation (EC) No. 1272/2008 [CLP/GHS]

Classification	Justification
Skin Sens. 1, H317	Calculation method
Aquatic Chronic 2, H411	Calculation method

Full text of abbreviated H statements	H304	May be fatal if swallowed and enters airways.
	H315	Causes skin irritation.
	H317	May cause an allergic skin reaction.
	H318	Causes serious eye damage.
	H361f	Suspected of damaging fertility.
	H361fd	Suspected of damaging fertility. Suspected of damaging the unborn child.
	H373	May cause damage to organs through prolonged or repeated exposure.
	H400	Very toxic to aquatic life.
	H410	Very toxic to aquatic life with long lasting effects.
	H411	Toxic to aquatic life with long lasting effects.
	H412	Harmful to aquatic life with long lasting effects.
	H413	May cause long lasting harmful effects to aquatic life.

Product name	SAF-PD	Product code	454252-DE01	Page:	21/28
Version	19	Date of issue	27 November 2025	Format	Germany
				Language	ENGLISH
Date of previous issue	20 February 2024.		(Germany)		

SECTION 16: Other information

Full text of classifications [CLP/GHS]	Aquatic Acute 1	SHORT-TERM (ACUTE) AQUATIC HAZARD - Category 1
	Aquatic Chronic 1	LONG-TERM (CHRONIC) AQUATIC HAZARD - Category 1
	Aquatic Chronic 2	LONG-TERM (CHRONIC) AQUATIC HAZARD - Category 2
	Aquatic Chronic 3	LONG-TERM (CHRONIC) AQUATIC HAZARD - Category 3
	Aquatic Chronic 4	LONG-TERM (CHRONIC) AQUATIC HAZARD - Category 4
	Asp. Tox. 1	ASPIRATION HAZARD - Category 1
	Eye Dam. 1	SERIOUS EYE DAMAGE/EYE IRRITATION - Category 1
	Repr. 2	REPRODUCTIVE TOXICITY - Category 2
	Skin Irrit. 2	SKIN CORROSION/IRRITATION - Category 2
	Skin Sens. 1	SKIN SENSITISATION - Category 1
	Skin Sens. 1B	SKIN SENSITISATION - Category 1B
	STOT RE 2	SPECIFIC TARGET ORGAN TOXICITY - REPEATED EXPOSURE - Category 2

History

Date of issue/ Date of revision	27/11/2025.
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Prepared by	Product Stewardship

Indicates information that has changed from previously issued version.

Notice to reader

All reasonably practicable steps have been taken to ensure this data sheet and the health, safety and environmental information contained in it is accurate as of the date specified below. No warranty or representation, express or implied is made as to the accuracy or completeness of the data and information in this data sheet.

The data and advice given apply when the product is sold for the stated application or applications. You should not use the product other than for the stated application or applications without seeking advice from BP Group.

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Annex to the extended Safety Data Sheet (eSDS)

Industrial

Identification of the substance or mixture

Product definition	Mixture
Code	454252-DE01
Product name	SAF-PD

Section 1: Title

Short title of the exposure scenario	General use of lubricants and greases in vehicles or machinery - Industrial
List of use descriptors	Identified use name: General use of lubricants and greases in vehicles or machinery-Industrial Process Category: PROC01, PROC02, PROC08b, PROC09 Sector of end use: SU03 Subsequent service life relevant for that use: No. Environmental Release Category: ERC04, ERC07 Specific Environmental Release Category: ATIEL-ATC SPERC 4.Biv1

Processes and activities covered by the exposure scenario	Covers general use of lubricants and greases in vehicles or machinery in closed systems. Includes filling and draining of containers and operation of enclosed machinery (including engines) and associated maintenance and storage activities.
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Section 2 Operational conditions and risk management measures

Section 2.1 Control of worker exposure

Product characteristics:

Physical state:	Liquid, vapour pressure < 0.5 kPa
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Concentration of substance in product:	Covers use of substance/product up to 100 % (unless stated differently)
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Frequency and duration of use:	Covers daily exposures up to 8 hours
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Other operational conditions affecting worker exposure:	Assumes use at not more than 20°C above ambient temperature, unless stated differently. Assumes a good basic standard of occupational hygiene is implemented
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Contributing scenarios: Operational conditions and risk management measures

General measures applicable to all activities:
 Avoid direct skin contact with product. Identify potential areas for indirect skin contact. Wear gloves (tested to EN 374) if hand contact with substance likely. Clean up contamination/spills as soon as they occur. Wash off any skin contamination immediately. Provide basic employee training to prevent/minimise exposures and to report any skin problems that may develop. Use suitable eye protection. Avoid direct eye contact with product also via contamination on hands.

General exposures (closed systems):
 No other specific measures identified.

Initial factory fill of equipment Use in contained systems:
 No other specific measures identified.

Initial factory fill of equipment Open systems:
 Provide a good standard of controlled ventilation (10 to 15 air changes per hour). Avoid carrying out operation for more than 4 hours.

Operation of equipment containing engine oils and similar Use in contained systems:
 No other specific measures identified.

Equipment cleaning and maintenance:
 Drain down system prior to equipment break-in or maintenance. Provide a good standard of general ventilation (not less than 3 to 5 air changes per hour). Wear chemically resistant gloves (tested to EN374) in combination with specific activity training. Retain drain-downs in sealed storage pending disposal or for subsequent recycle.

Equipment cleaning and maintenance Operation is carried out at elevated temperature (> 20°C above ambient temperature):
 Drain down and flush system prior to equipment break-in or maintenance. Provide extract ventilation to emission

SAF-PD

General use of lubricants and greases in vehicles or machinery - Industrial

points when contact with warm (>50°C) lubricant is likely. Wear chemically resistant gloves (tested to EN374) in combination with intensive management supervision controls. Retain drain-downs in sealed storage pending disposal or for subsequent recycle.

Storage:
Store substance within a closed system.

Section 2.2: Control of environmental exposure

Amounts used:

EU tonnage of risk determining substance per year: 2.63E+3 Tonnes/year

Frequency and duration of use:

Emission days 300

Environment factors not influenced by risk management:

Local freshwater dilution factor 10

Local marine water dilution factor 100

Other operational conditions of use affecting environmental exposure: Negligible wastewater emissions as process operates without water contact.

Release fraction to wastewater from process (after typical onsite RMMs and before sewage treatment plan) 2.00E-11

Technical conditions and measures at process level (source) to prevent release:

Common practices vary across sites thus conservative process release estimates used.

Technical on-site conditions and measures to reduce or limit discharges, air emissions and releases to soil:

Prevent discharge of undissolved substance to or recover from onsite wastewater.
User sites are assumed to be provided with oil/water separators and waste water to be discharged via a sewage treatment plant

Organisational measures to prevent/limit release from site:

Do not apply industrial sludge to natural soils.
Sludge should be incinerated, contained or reclaimed.

Conditions and measures related to municipal sewage treatment plant:

Estimated substance removal from wastewater via on-site sewage treatment 79

Assumed domestic sewage treatment plant flow rate (m3/d) 2.00E+3

Maximum allowable site tonnage (M_{Safe}) based on release following total wastewater treatment removal as product: 208241.12

Conditions and measures related to external treatment of waste for disposal:

External treatment and disposal of waste should comply with applicable local and/or national regulations.

Conditions and measures related to external recovery of waste:

External recovery and recycling of waste should comply with applicable local and/or national regulations.

Section 3: Exposure estimation and reference to its source

Exposure estimation and reference to its source - Environment

Exposure assessment (environment): Used ECETOC TRA model (May 2010 release).

Exposure estimation and reference to its source - Workers

Exposure assessment (human): The ECETOC TRA tool has been used to estimate workplace exposures unless otherwise indicated.

Section 4: Guidance to check compliance with the exposure scenario

Environment

Guidance is based on assumed operating conditions which may not be applicable to all sites; thus, scaling may be necessary to define appropriate site-specific risk management measures. Further details on scaling and control technologies are provided in SPERC factsheet. If scaling reveals a condition of unsafe use (i.e., RCRs > 1), additional RMMs or a site-specific chemical safety assessment is required. For further information see www.ATIEL.org/REACH_GES

Health

Where other risk management measures/operational conditions are adopted, then users should ensure that risks are managed to at least equivalent levels.

Annex to the extended Safety Data Sheet (eSDS)

Professional

Identification of the substance or mixture

Product definition	Mixture
Code	454252-DE01
Product name	SAF-PD

Section 1: Title

Short title of the exposure scenario	General use of lubricants and greases in vehicles or machinery - Professional
List of use descriptors	Identified use name: General use of lubricants and greases in vehicles or machinery-Professional Process Category: PROC01, PROC02, PROC08b, PROC08a, PROC20 Sector of end use: SU22 Subsequent service life relevant for that use: No. Environmental Release Category: ERC09a, ERC09b Specific Environmental Release Category: ATIEL-ATC SPERC 4.Biv1

Processes and activities covered by the exposure scenario	Covers general use of lubricants and greases in vehicles or machinery in closed systems. Includes filling and draining of containers and operation of enclosed machinery (including engines) and associated maintenance and storage activities.
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Section 2 Operational conditions and risk management measures

Section 2.1 Control of worker exposure

Product characteristics:

Physical state:	Liquid, vapour pressure < 0.5 kPa
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Concentration of substance in product:	Covers use of substance/product up to 100 % (unless stated differently)
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Frequency and duration of use:	Covers daily exposures up to 8 hours
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Other operational conditions affecting worker exposure:	Assumes use at not more than 20°C above ambient temperature, unless stated differently. Assumes a good basic standard of occupational hygiene is implemented
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Contributing scenarios: Operational conditions and risk management measures

General measures applicable to all activities:
 Avoid direct skin contact with product. Identify potential areas for indirect skin contact. Wear gloves (tested to EN 374) if hand contact with substance likely. Clean up contamination/spills as soon as they occur. Wash off any skin contamination immediately. Provide basic employee training to prevent/minimise exposures and to report any skin problems that may develop. Use suitable eye protection. Avoid direct eye contact with product also via contamination on hands.

General exposures (closed systems):
 No other specific measures identified.

Initial factory fill of equipment Use in contained systems:
 No other specific measures identified.

Initial factory fill of equipment Open systems:
 Provide a good standard of controlled ventilation (10 to 15 air changes per hour). Avoid carrying out operation for more than 4 hours.

Operation of equipment containing engine oils and similar Use in contained systems:
 No other specific measures identified.

Equipment cleaning and maintenance:
 Drain down system prior to equipment break-in or maintenance. Provide a good standard of general ventilation (not less than 3 to 5 air changes per hour). Wear chemically resistant gloves (tested to EN374) in combination with specific activity training. Retain drain-downs in sealed storage pending disposal or for subsequent recycle.

Equipment cleaning and maintenance Operation is carried out at elevated temperature (> 20°C above ambient temperature):
 Drain down and flush system prior to equipment break-in or maintenance. Provide extract ventilation to emission

SAF-PD

General use of lubricants and greases in vehicles or machinery - Professional

points when contact with warm (>50°C) lubricant is likely. Wear chemically resistant gloves (tested to EN374) in combination with intensive management supervision controls. Retain drain-downs in sealed storage pending disposal or for subsequent recycle.

Storage:

Store substance within a closed system.

Section 2.2: Control of environmental exposure

Amounts used:

EU tonnage of risk determining substance per year: 2.63E+3 Tonnes/year

Frequency and duration of use:

Emission days 300

Environment factors not influenced by risk management:

Local freshwater dilution factor 10

Local marine water dilution factor 100

Other operational conditions of use affecting environmental exposure: Negligible wastewater emissions as process operates without water contact.

Release fraction to wastewater from process (after typical onsite RMMs and before sewage treatment plan) 5.00E-04

Technical conditions and measures at process level (source) to prevent release:

Common practices vary across sites thus conservative process release estimates used.

Technical on-site conditions and measures to reduce or limit discharges, air emissions and releases to soil:

Prevent discharge of undissolved substance to or recover from onsite wastewater.
User sites are assumed to be provided with oil/water separators and waste water to be discharged via a sewage treatment plant

Organisational measures to prevent/limit release from site:

Do not apply industrial sludge to natural soils.
Sludge should be incinerated, contained or reclaimed.

Conditions and measures related to municipal sewage treatment plant:

Estimated substance removal from wastewater via on-site sewage treatment 79

Assumed domestic sewage treatment plant flow rate (m3/d) 2.00E+3

Maximum allowable site tonnage (M_{Safe}) based on release following total wastewater treatment removal as product: 1.6

Conditions and measures related to external treatment of waste for disposal:

External treatment and disposal of waste should comply with applicable local and/or national regulations.

Conditions and measures related to external recovery of waste:

External recovery and recycling of waste should comply with applicable local and/or national regulations.

Section 3: Exposure estimation and reference to its source

Exposure estimation and reference to its source - Environment

Exposure assessment (environment): Used ECETOC TRA model (May 2010 release).

Exposure estimation and reference to its source - Workers

Exposure assessment (human): The ECETOC TRA tool has been used to estimate workplace exposures unless otherwise indicated.

Section 4: Guidance to check compliance with the exposure scenario

Environment

Guidance is based on assumed operating conditions which may not be applicable to all sites; thus, scaling may be necessary to define appropriate site-specific risk management measures. Further details on scaling and control technologies are provided in SPERC factsheet. If scaling reveals a condition of unsafe use (i.e., RCRs > 1), additional RMMs or a site-specific chemical safety assessment is required. For further information see www.ATIEL.org/REACH_GES

Health

Where other risk management measures/operational conditions are adopted, then users should ensure that risks are managed to at least equivalent levels.